# Mitigating Anchoring Bias in Long-Term Auditor Engagements

**A Pilot and Feasibility Study**

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# CHAPTER 1 — INTRODUCTION

## 1.1 Background of the Study

Auditing is, at its core, a discipline of professional judgment exercised under uncertainty. Auditors form expectations about account balances, estimates, and risks, then test those expectations against evidence. Decades of research in judgment and decision-making have established that such expectation-formation is systematically vulnerable to cognitive heuristics — and among these, **anchoring** is one of the most robust: when a judgment begins from an initial reference point, subsequent adjustment away from that reference point is typically insufficient, leaving the final judgment biased toward the anchor (Tversky & Kahneman, 1974). In the audit context, anchors are structural, not incidental. Prior-year balances, management's estimates, budgets, and prior working papers are embedded in the audit process itself; the auditor does not need to seek an anchor, because the engagement file supplies one.

This vulnerability is sharpest in **long-term, continuing engagements** — multi-year relationships in which the same audit team returns to the same client, carrying forward the same working papers and conclusions. Recent experimental evidence with practicing auditors confirms that internally generated anchors such as prior-year working papers exert the strongest pull on audit judgments, and that susceptibility varies with experience and with the auditor's awareness of behavioral heuristics (Henrizi, Himmelsbach, & Hunziker, 2021). What begins as engagement efficiency — leveraging prior knowledge of the client — can quietly become engagement risk: an expectation inherited rather than formed, tested against evidence chosen in its own shadow.

The profession is not defenseless. Audit firms and standard-setters have developed an arsenal of practices with the potential to disrupt anchoring: bias-awareness training, rotation of engagement personnel, data-analytical tools that generate independent expectations, structured audit processes and decision aids, feedback and reflection practices, independent reviews, regulatory and professional guidance, and performance-evaluation systems that reward judgment quality over speed. Yet these interventions have largely been studied piecemeal. What the literature lacks — and what this study set out to build — is an integrated measurement framework that captures these interventions, the judgment conditions they operate through, and the outcome they target, in a single validated instrument grounded in the experience of practicing auditors.

## 1.2 Problem Statement

Anchoring bias in long-term auditor engagements threatens audit quality at precisely the point where audit failures are most costly: the evaluation of recurring balances and estimates against inherited expectations. Although individual mitigation practices are known, there is no validated, integrated measurement model of the organizational interventions that reduce anchoring in continuing engagements, nor of the mechanisms — the quality of auditor judgment and the rigor of the audit process — through which those interventions plausibly operate. Without validated measurement, firms cannot assess which of their quality-control investments actually reach the judgment behaviors that matter, and researchers cannot test intervention effectiveness at scale. The problem this study addresses is therefore a **measurement problem with practical stakes**: the absence of a validated instrument linking anchoring-mitigation interventions to reduced anchoring bias in the professionals most exposed to it.

## 1.3 Purpose of the Study

The purpose of this study was to develop and begin validating an eleven-construct measurement model of anchoring-bias mitigation in long-term auditor engagements. The model comprises eight organizational interventions (Training & Awareness; Rotation of Auditors; Use of Analytical Tools; Structured Auditing Processes; Feedback & Reflection; Independent Reviews; Regulatory & Professional Guidance; Performance Metrics & Incentives), two mediating conditions (Auditor Judgment Quality; Audit Process Rigor), and one outcome construct (Reduction in Anchoring Bias), operationalized as a 55-item survey instrument administered to practicing U.S. auditors with continuing-engagement experience. The empirical scope of the present study, set in consultation with the faculty advisor, was the measurement stage: exploratory factor analysis and reliability assessment of the instrument, with the full structural model — sixteen hypothesized relationships across direct and mediated paths — specified as the framework for subsequent testing.

As Chapters 5 and 6 report candidly, the study's data-collection effort encountered a structural constraint — the extreme narrowness of the eligible population — that converted the planned validation study into a **pilot and feasibility study**. The purpose statement above therefore describes both what the study set out to do and the apparatus it delivers intact for the next stage of the research program; what changed in execution, and what was learned from that change, is itself a central contribution of this paper.

## 1.4 Research Questions

The study is organized around three questions:

- **RQ1.** Can the eight anchoring-mitigation interventions, two mediating conditions, and the anchoring-reduction outcome be measured as distinct, reliable constructs among practicing auditors in long-term engagements?
- **RQ2.** What is the underlying factor structure of the 55-item instrument — do the items cohere into the eleven theorized constructs?
- **RQ3.** As the framework for subsequent research: how are the interventions associated with reduced anchoring bias, directly and through auditor judgment quality and audit process rigor?

The present study's empirical work addresses RQ1 and RQ2 to the extent the achieved sample permits; RQ3 is developed theoretically in Chapter 3 and reserved, with its sixteen hypotheses, for the adequately powered study this paper prepares.

## 1.5 Significance of the Study

The study matters to three audiences. **For audit practice**, a validated instrument would let firms measure — rather than assume — whether their training, rotation, analytics, review, and incentive structures actually reach anchoring behavior in continuing engagements, converting quality-control spending into testable interventions. **For the research literature**, the study integrates intervention constructs that have been examined only in isolation, and grounds them in a dual-process theoretical frame with explicit mediating mechanisms. **For the emerging technology agenda**, the framework arrives as the profession absorbs AI- and LLM-based audit tools, where the central open question — developed in Chapter 6 — is whether algorithmic decision aids attenuate human anchors or substitute algorithmic anchors of their own (Commerford, Dennis, Joe, & Ulla, 2022; Kokina, Blanchette, Davenport, & Pachamanova, 2025). A validated human-side baseline is the precondition for answering that question, and this study builds it. Finally, the study's candid documentation of its data-collection experience — the true cost and feasibility of reaching a narrow professional population — is itself significant for doctoral researchers designing studies of specialized populations.

## 1.6 Definitions of Key Terms

- **Anchoring bias** — the tendency for judgments to remain biased toward an initial reference value (prior-year balance, management estimate, budget) after insufficient adjustment (Tversky & Kahneman, 1974).
- **Long-term (continuing) engagement** — a multi-year audit relationship in which engagement personnel, working papers, and conclusions carry forward across periods.
- **Anchoring-mitigation interventions** — the eight organizational practices theorized to reduce anchoring: training and awareness, auditor rotation, analytical tools, structured processes, feedback and reflection, independent reviews, regulatory and professional guidance, and performance metrics and incentives.
- **Auditor Judgment Quality (AJQ)** — the degree to which audit judgments rest on careful, objective evaluation of independent evidence (cognitive mediator).
- **Audit Process Rigor (APR)** — the thoroughness, consistency, and disciplined evidence-ordering of the audit process as executed (procedural mediator).
- **Reduction in Anchoring Bias (RAB)** — the outcome construct: the extent to which final judgments are driven by current evidence rather than initial reference points.
- **Pilot / feasibility study** — a study whose achieved scope is the demonstration that the instrument, procedures, and analytic pipeline function as designed, and the identification of the constraints a full-scale study must solve.

## 1.7 Organization of the Study

Chapter 2 reviews the literature: the theoretical foundations, each of the eleven constructs, and the research gap. Chapter 3 presents the research model and develops the sixteen hypotheses that constitute the study's structural framework. Chapter 4 specifies the complete methodology as an analysis contract — design, instrument, sampling plan, data-preparation protocol, and the full analytic sequence from exploratory factor analysis through reliability, confirmatory factor analysis, regression, and PLS-SEM, with the decision criteria governing each stage. Chapter 5 documents the data-collection experience and its outcome, and states what the achieved sample did and did not permit. Chapter 6 draws conclusions: the study's limitations, recommendations for researchers facing comparable populations, and the transition from this qualifying study to the dissertation, including its extension into AI-augmented audit judgment.

# CHAPTER 2 — REVIEW OF THE LITERATURE

## 2.1 Introduction to the Chapter

This chapter reviews the body of scholarship on which the present study is built. It has three purposes. First, it introduces the theoretical foundations that explain why anchoring bias arises in human judgment generally and in auditor judgment specifically, and it identifies the single theoretical lens that organizes the remainder of the study. Second, it examines, one construct at a time, each of the eleven concepts that compose the research model — the eight audit-process interventions, the two judgment-process mediators, and the dependent variable — summarizing what the literature has established about each, where scholars disagree, and what remains unresolved. Third, it synthesizes these separate literatures into the research gap that motivates the integrated model developed in Chapter 3.

The chapter is deliberately organized so that the constructs are treated here as *separate* bodies of knowledge: each section describes a single "box" in the eventual model, the way the literature defines it, measures it, and links it to professional-judgment quality. The relationships *among* the boxes — the arrows, the directional hypotheses, and the mediating pathways — are reserved for Chapter 3. This separation follows the convention that a literature review establishes what is known about each concept, while the model-and-hypotheses chapter argues for the specific causal relationships the study will test (Bonner, 2008; Nelson & Tan, 2005).

Anchoring bias is among the most robust and widely replicated findings in the judgment-and-decision-making literature (Furnham & Boo, 2011; Tversky & Kahneman, 1974). In the audit setting it is also among the most consequential, because auditors who return year after year to the same long-term client are repeatedly exposed to a powerful set of anchors — prior-period balances, management's estimates, last year's workpapers, and the client's own explanations. When these anchors are weighted too heavily, the auditor's adjustment toward an independent, evidence-based conclusion is insufficient, and professional skepticism is quietly eroded (Joyce & Biddle, 1981; Kinney & Uecker, 1982; Nelson, 2009). The literature reviewed below establishes both the depth of this problem and the fragmented state of the interventions proposed to address it.

## 2.2 Theoretical Foundations

A theory of anchoring is required before the model's constructs can be justified, because each intervention in the model is ultimately an attempt to interrupt a cognitive mechanism that theory specifies. Anchoring bias has been explained by several complementary theoretical accounts. In the judgment-and-decision-making and audit literatures, the most influential are: (a) the **anchoring-and-adjustment heuristic** (Tversky & Kahneman, 1974; Epley & Gilovich, 2006); (b) the **selective-accessibility, or semantic-priming, account** (Strack & Mussweiler, 1997); (c) **dual-process theory**, which distinguishes fast, automatic System-1 cognition from slow, deliberate System-2 reasoning (Evans, 2008; Kahneman, 2011); (d) the **belief-adjustment model** of sequential evidence evaluation (Hogarth & Einhorn, 1992); and (e) the **audit-quality and judgment-performance frameworks** that connect process attributes to judgment outcomes (Knechel, 2013; Libby & Luft, 1993; Nelson & Tan, 2005). Each is introduced in turn below for a reader encountering it for the first time, after which the study's selected lens is named and justified.

### 2.2.1 The Anchoring-and-Adjustment Heuristic

Tversky and Kahneman (1974) introduced anchoring as one of three general heuristics people use to reduce the complexity of judgment under uncertainty. In their account, individuals form estimates by starting from an initial value — the anchor — and adjusting from it to reach a final answer; critically, the adjustment is typically insufficient, so the final estimate remains biased toward the anchor. The effect proved extraordinarily robust: it appears even when the anchor is known to be random, uninformative, or implausibly extreme, and even when respondents are warned about it and paid for accuracy (Furnham & Boo, 2011). Wilson, Houston, Etling, and Brekke (1996), in the *Journal of Experimental Psychology: General*, demonstrated "basic anchoring" — that uninformative numerical anchors influence subsequent judgments even when no explicit comparison to the target is requested, expanding the conditions under which anchoring is now known to operate. Epley and Gilovich (2006) refined the mechanism, showing that anchoring actually reflects (at least) two processes — insufficient *effortful* adjustment from self-generated anchors and a more automatic priming effect from externally provided anchors — a distinction that matters for debiasing, because the two are interrupted by different interventions. For the audit setting, the heuristic's relevance is direct: prior-year balances and management estimates function as externally provided anchors from which the auditor adjusts too little, and Northcraft and Neale (1987), in *Organizational Behavior and Human Decision Processes*, supplied an early demonstration that experts in an information-rich professional setting (real-estate appraisers) anchor on listing prices much as novices do — undermining the assumption that domain expertise alone neutralizes the bias.

### 2.2.2 The Selective-Accessibility (Semantic-Priming) Account

Strack and Mussweiler (1997) proposed that anchoring is, in large part, a special case of semantic priming. On their account, considering an anchor causes the judge to selectively retrieve and activate information *consistent* with that anchor; this anchor-consistent information is then disproportionately accessible when the absolute judgment is made, biasing it toward the anchor. Mussweiler and Strack (1999), in the *Journal of Experimental Social Psychology*, extended the account by showing that the mechanism is *hypothesis-consistent testing*: judges implicitly test the hypothesis that the anchor is the correct value, which selectively raises the accessibility of anchor-consistent knowledge used in the subsequent judgment. The selective-accessibility model explains why anchoring is so resistant to simple warnings and why a particularly effective countermeasure is the deliberate "consider-the-opposite" strategy (Lord, Lepper, & Preston, 1984; Mussweiler, Strack, & Pfeiffer, 2000), in which the judge is prompted to generate anchor-*inconsistent* evidence. This account is the theoretical bridge to several interventions in the present model — independent review, structured procedures, and feedback all function, in part, by forcing the auditor to surface evidence inconsistent with the prior-year anchor.

### 2.2.3 Dual-Process Theory of Judgment

Dual-process theory holds that cognition operates through two interacting systems: System 1 is fast, automatic, associative, and effortless; System 2 is slow, controlled, rule-governed, and effortful (Evans, 2008; Kahneman, 2011). Kahneman and Frederick (2002), in their "representativeness revisited" chapter, formalized the dual-process account through the concept of *attribute substitution* — when a target attribute is hard to assess, individuals unwittingly substitute a related but easier-to-evaluate attribute — providing a unifying mechanism that links anchoring, representativeness, and availability under the same theoretical umbrella. Anchoring is understood as a System-1 phenomenon — the anchor exerts its pull automatically — that can be overridden by System-2 engagement when, and only when, the judge has the motivation, the cognitive resources, and a structure that prompts deliberate reconsideration. In auditing, Nelson

(2009) and Libby and Luft (1993) frame professional skepticism and judgment performance in compatible terms: knowledge, ability, motivation, and environment jointly determine whether the deliberate, evidence-driven system is actually deployed. Dual-process theory is thus the natural framework for an *intervention* study, because every construct in the model can be read as a lever that increases the probability that effortful System-2 reasoning displaces automatic System-1 anchoring.

#### 2.2.4 The Belief-Adjustment Model

Hogarth and Einhorn (1992) developed the belief-adjustment model to explain how judgments evolve as evidence is processed in sequence. Their model is itself a generalized anchoring-and-adjustment process: a current belief serves as the anchor, and each new evidence item produces an adjustment whose size depends on task complexity, the length of the evidence series, and whether evidence is processed step-by-step or end-of-sequence. The model predicts systematic order effects — notably recency in step-by-step processing of mixed evidence — that are directly relevant to multi-year audit engagements, where the order and framing of evidence (last year's conclusion first, current evidence second) can entrench the prior-period anchor. The belief-adjustment model supplies the theoretical rationale for interventions that alter *how* and *in what order* evidence is evaluated, such as structured processes and analytical tools.

#### 2.2.5 Audit-Quality and Judgment-Performance Frameworks

Finally, a stream of audit-specific frameworks connects the attributes of the audit *process* to the quality of audit *judgments* and, ultimately, to audit quality (DeFond & Francis, 2005; Knechel, 2013; Nelson & Tan, 2005). Libby and Luft's (1993) determinants-of-performance framework — ability, knowledge, motivation, and environment — is the canonical organizing scheme, and Bonner (2008) provides the comprehensive treatment of judgment-and-decision-making in accounting on which most measurement in this area rests. These frameworks do not, by themselves, explain anchoring; rather, they specify the conditions under which any debiasing intervention will or will not improve judgment, and they supply the validated construct definitions and measurement logic that the present study adopts.

#### 2.2.6 Selected Theoretical Lens for the Present Study

While all five accounts inform this study, the research adopts **dual-process theory, operationalized through the anchoring-and-adjustment heuristic, as its primary theoretical lens.** This choice is deliberate and is defended on three grounds. First, it is the most *integrative* of the available theories: it subsumes the anchoring-and-adjustment heuristic (anchoring as System-1) and is fully compatible with both the selective-accessibility and belief-adjustment accounts (each describes a System-1 mechanism that System-2 engagement can correct). Second, it is the most *actionable* lens for an intervention study, because every construct in the model can be expressed as a mechanism that raises the likelihood, capacity, or required structure for System-2 reasoning to override the anchor — making the theory directly testable through the hypotheses in Chapter 3. Third, it is the lens most consistent with how the *audit* literature already theorizes professional skepticism and judgment quality (Nelson, 2009; Nelson & Tan, 2005), which preserves continuity with the validated instruments the study adapts. The selective-accessibility, belief-adjustment, and audit-quality frameworks are retained as supporting theory wherever a specific construct's mechanism is better described by them.

### 2.3 Reduction in Anchoring Bias (RAB) — The Dependent Variable

The dependent variable, Reduction in Anchoring Bias (RAB), is defined as the extent to which an auditor's judgments are *free* from the distorting influence of an initial reference point — that is, the degree to which the auditor adjusts adequately away from prior-period balances, management estimates, and previously documented conclusions toward an independent, evidence-based position (Tversky & Kahneman, 1974). Anchoring is operationalized in the behavioral-auditing literature as the residual pull of an anchor on a final judgment, and a *reduction* in that pull is the outcome every intervention in this model is theorized to produce.

The audit-specific evidence that anchoring genuinely afflicts professional judgment is direct and longstanding. Joyce and Biddle (1981), in a set of experiments with practicing auditors published in the *Journal of Accounting Research*, found that auditors' probabilistic inferences were influenced by anchors, though not always in the simple direction a naive heuristic account predicts — establishing both the presence of the effect and its complexity in expert populations. Kinney and Uecker (1982), in *The Accounting Review*, demonstrated anchoring and adjustment in analytical review: when auditors were given biased unaudited figures, those figures shifted the investigation boundaries the auditors set, showing that an anchor can distort not only an estimate but the *scope of testing* itself. Wilson et al. (1996) and Northcraft and Neale (1987) generalized the concern to expert populations more broadly, showing that uninformative anchors influence judgments even of professionals operating in their home domains. Furnham and Boo (2011) review the broader literature and document the effect's remarkable robustness across domains, populations, and incentives — a robustness that is precisely why the audit profession cannot assume expertise alone neutralizes it.

A central tension in this literature is whether expertise and accountability attenuate anchoring. Some studies find that experienced auditors are less susceptible than novices, while others find professionals fully susceptible; Kennedy (1993) shows that accountability reduces some judgment biases only when the bias originates in insufficient effort rather than in data or memory limitations — implying that anchoring is reducible only by interventions matched to its actual cognitive source. This unresolved boundary condition is the gap the present model addresses by testing eight interventions simultaneously rather than in isolation.

## 2.4 Training & Awareness (TA)

Training & Awareness (TA) refers to structured instruction designed to make auditors aware of cognitive biases — anchoring among them — and to equip them with concrete debiasing strategies they can deploy during fieldwork. Earley (2001), in *Accounting Horizons*, showed that knowledge acquisition in auditing is most effective when learning is structured and accompanied by explanatory feedback, producing measurable gains in judgment performance among less-experienced auditors. Bonner and Walker (1994), in *The Accounting Review*, demonstrated that the *combination* of instruction with practice-and-feedback — not instruction alone — was what produced durable procedural knowledge for analytical tasks, a finding that distinguishes genuine debiasing training from mere awareness. Hurtt, Brown-Liburd, Earley, and Krishnamoorthy (2013), synthesizing the auditor-skepticism literature in *Auditing: A Journal of Practice & Theory*, identify training and experience as among the most studied antecedents of professional skepticism — the auditor mindset most directly implicated in resisting anchors — and document that the evidence on training's effect is positive but heterogeneous across study designs.

The literature is, however, divided on whether awareness translates into behavior. A recurring finding in the broader debiasing literature is that simply *informing* people about a bias rarely removes it, because anchoring operates automatically (Furnham & Boo, 2011; Strack & Mussweiler, 1997). Solomon and Trotman (2003), in their *Accounting, Organizations and Society* review of twenty-five years of auditing judgment research, similarly note that interventions targeting knowledge structures are more effective when they alter how auditors *process* evidence rather than merely what they *know* about biases. This is the central contradiction the TA construct must confront: awareness is necessary but not sufficient, and training appears to "work" only when it installs an executable strategy (e.g., consider-the-opposite, independent estimation before reviewing prior-year figures) rather than abstract knowledge that a bias exists. The most defensible reading of the literature is therefore that TA reduces anchoring to the degree that it changes the auditor's *process*, not merely the auditor's *beliefs*.

## 2.5 Rotation of Auditors (RA)

Rotation of Auditors (RA) refers to the periodic reassignment of audit personnel (at the engagement, partner, or firm level) intended to disrupt the relational and informational lock-in that accumulates over a long-term engagement. The rationale is that a fresh auditor carries no prior-year anchor of their own and no relationship-based deference to management. Arel, Brody, and Pany (2005) summarize the policy logic for rotation and audit quality, while Chi, Huang, Liao, and Xie (2009), in *Contemporary Accounting Research*, examined mandatory partner rotation and audit quality using archival data, and Chen, Lin, and Lin (2016) studied audit-partner tenure and discretionary accruals — together mapping how the length of the auditor–client relationship relates to reporting outcomes.

The archival evidence on tenure is, however, decidedly mixed, and that mixed pattern is itself diagnostic. Carcello and Nagy (2004), in *Auditing: A Journal of Practice & Theory*, found that fraudulent financial reporting is more likely in the *first three years* of an auditor–client relationship and find no evidence that long tenure increases fraud risk — implying that start-up unfamiliarity, not familiarity, is the more proximate threat. Myers, Myers, and Omer (2003), in *The Accounting Review*, similarly report that longer auditor tenure is associated with *higher* earnings quality (smaller absolute abnormal accruals), again challenging the assumption that long tenure mechanically degrades audit quality. The experimental evidence sharpens the picture: Bowlin, Hobson, and Piercey (2015), in *The Accounting Review*, found that mandatory rotation can *improve* audit quality when auditors adopt an honesty frame but can *reduce* audit effort when auditors already hold a skeptical frame — because rotating auditors, anticipating a short relationship, may invest less in understanding a new client. Rotation thus interacts with the very skepticism it is meant to protect, and the field-level archival evidence shows that simple "tenure-is-bad" intuitions are unsupported. The literature therefore does not support a simple "rotation reduces bias" claim; it supports the more careful position that rotation removes specific *relationship-based* anchors while introducing its own costs (lost client-specific knowledge, start-up inefficiency), and the net direction is an empirical question this model is designed to estimate.

## 2.6 Use of Analytical Tools (AT)

Use of Analytical Tools (AT) refers to data-analytics, visualization, and anomaly-detection technologies that surface patterns and outliers in client data, redirecting the auditor's attention from historical expectations toward current evidence. Yoon, Hoogduin, and Zhang (2015), in *Accounting Horizons*, argued that big-data sources can serve as complementary audit evidence, expanding the evidence base beyond the client's own ledgers. Brazel, Jones, and Prawitt (2014) showed that attention to non-financial measures that are inconsistent with reported financials improves fraud-risk judgments — a mechanism that directly counteracts anchoring, because it confronts the auditor with disconfirming data. Trompeter and Wright (2010), in *Contemporary Accounting Research*, documented how analytical-procedure *practice* has shifted with technology toward more precise, quantitative, and externally benchmarked expectations. The longer arc of this transition is captured by Vasarhelyi, Alles, and Kogan (2004), whose foundational treatment of continuous auditing in the *Journal of Emerging Technologies in Accounting* sets out how information-technology-enabled monitoring can convert episodic, retrospective testing into ongoing analytic comparison — structurally raising the salience of current-period anomalies relative to prior-year expectations.

The tension in this literature concerns reliance and automation. Asare and Wright (2004), in *Contemporary Accounting Research*, found that the design of risk-assessment and program-planning tools materially shapes auditor judgments — meaning a poorly designed tool can *introduce* an anchor (e.g., a default risk score) as readily as a good tool removes one. Analytical tools, in other words, are debiasing only insofar as they direct attention to anchor-inconsistent evidence; when they present a salient default or prior-year benchmark, they risk re-anchoring the auditor. This double-edged quality is why AT is modeled as an intervention whose effect must be estimated empirically rather than assumed.

## 2.7 Structured Auditing Processes (SAP)

Structured Auditing Processes (SAP) refers to standardized procedures, checklists, decision aids, and documentation templates that prescribe the sequence and content of audit work, replacing ad hoc heuristic shortcuts with explicit, repeatable steps. Dowling and Leech (2007), in the *International Journal of Accounting Information Systems*, analyzed audit-support systems and decision aids and how their structure shapes auditor behavior and compliance. Bonner (2008) situates structure within the broader judgment-and-decision-making literature, noting that imposed structure can substitute for missing knowledge and force consideration of factors a less-structured auditor might skip. Carpenter (2007), in *The Accounting Review*, provided causal evidence for one form of imposed structure: SAS No. 99 brainstorming sessions raised the quality of fraud-risk ideas generated by audit teams above what individual auditors produced alone, and team fraud-risk assessments after brainstorming were significantly higher — demonstrating that a structured group procedure can override the prior-year frame an individual auditor would otherwise default to. Trotman, Bauer, and Humphreys (2015), reviewing group judgment-and-decision-making research in auditing in *Accounting, Organizations and Society*, situate brainstorming, hierarchical review, and consultation as the three principal structured processes through which audit teams produce judgments that exceed any single member's, and they highlight the importance of *process design* in whether these benefits are realized.

The literature's enduring debate is the "structure versus judgment" trade-off. Structure can suppress anchoring by mandating independent evidence evaluation before the prior-year conclusion is consulted, and by requiring documentation that exposes insufficient adjustment. Yet over-structuring can mechanize the audit, encouraging compliance-as-checkbox and *reducing* the very System-2 engagement that overrides anchors — the auditor follows the template without thinking. The defensible synthesis, consistent with the belief-adjustment model (Hogarth & Einhorn, 1992), is that structure reduces anchoring when it alters the *order and independence* of evidence evaluation, but not when it merely adds documentation around an unchanged thought process. This makes SAP a procedural-rigor lever, which is why Chapter 3 routes it through the Audit Process Rigor mediator.

## 2.8 Feedback & Reflection (FR)

Feedback & Reflection (FR) refers to metacognitive mechanisms — post-task debriefs, outcome and process feedback, and structured reflection — through which auditors learn to recognize and correct their own judgment tendencies. Bonner and Walker (1994) established that feedback (especially explanatory, process-oriented feedback) is a necessary ingredient for acquiring durable audit knowledge, and Bamber and Iyer (2007), in *Auditing: A Journal of Practice & Theory*, examined how auditors' identification with clients affects objectivity — a tendency that reflection is meant to counteract. Kennedy (1993) provides the most direct evidence relevant to anchoring: holding auditors accountable for their *process* reduces effort-related biases, implying that reflection works by raising effortful System-2 engagement. The cognitive-psychology evidence reinforces the mechanism: Lord, Lepper, and Preston (1984), in the *Journal of Personality and Social Psychology*, showed that inducing judges to *consider the opposite* of their initial impression is among the most reliably effective debiasing strategies known, and Mussweiler and Strack (1999) tied this finding directly to anchoring by showing that the bias is mediated by hypothesis-consistent testing — precisely the cognitive habit that structured reflection is designed to interrupt.

The key qualification in this literature is that feedback only debiases when the environment provides it in a usable form. Audit settings are notorious for delayed, noisy, or absent outcome feedback (the "true" answer often never arrives), which is why the literature emphasizes *process* feedback and structured reflection over outcome feedback (Bonner, 2008; Kennedy, 1993). The contradiction to manage is that reflection can also rationalize a prior-year anchor after the fact rather than dislodge it; feedback is debiasing only when it is structured to compare the auditor's judgment against independent evidence, not against the anchor itself.

## 2.9 Independent Reviews (IR)

Independent Reviews (IR) refers to the examination of audit work by a second, uninvolved auditor — engagement-quality reviewers, peer reviewers, or quality-assurance functions — who did not form the original judgment and therefore does not share the original anchor. DeFond and Francis (2005), in *Auditing: A Journal of Practice & Theory*, frame review and quality control within the post-Sarbanes-Oxley audit-quality architecture. The foundational behavioral evidence is Trotman and Yetton (1985), in the *Journal of Accounting Research*, who showed that the review process improves audit judgments, with reviewed judgments outperforming individual ones. Ramsay (1994), also in *JAR*, refined this by showing that reviewers at different ranks detect different kinds of errors — managers catching conceptual errors that seniors miss — establishing that review quality depends on reviewer expertise and focus. Tan and Trotman (2003), in *The Accounting Review*, sharpened the construct further by demonstrating that reviewers' judgments are themselves influenced by the *structure and conclusions* documented in the preparer's workpapers — meaning the independence of the reviewer's contribution is contingent on the format in which the preparer's work is presented. Solomon and Trotman (2003), reviewing twenty-five years of *Accounting, Organizations and Society* judgment research, situate the review process as the audit profession's primary structural mechanism for upgrading individual judgment to team judgment.

The selective-accessibility account (Strack & Mussweiler, 1997) explains *why* independent review is theoretically among the strongest anti-anchoring interventions: a reviewer who did not generate the original judgment is not primed by the same anchor and can surface anchor-inconsistent evidence. The contradiction the literature flags is that review loses its independence advantage when the reviewer is exposed to, and defers to, the preparer's documented conclusion before forming an independent view — re-creating the anchor at the review level (Tan & Trotman, 2003). Review is therefore debiasing in proportion to its genuine *independence*, which is why it is modeled (with SAP) as a driver of Audit Process Rigor rather than a purely cognitive lever.

## 2.10 Regulatory & Professional Guidance (RPG)

Regulatory & Professional Guidance (RPG) refers to the standards, inspection regimes, and professional requirements — PCAOB standards and inspections, professional skepticism mandates, and firm methodology aligned to them — that constrain auditor discretion and prescribe minimum judgment quality. Knechel (2013), in *Current Issues in Auditing*, asks directly whether auditing standards matter for judgment and quality, and the PCAOB's standards (PCAOB, 2015) codify the requirements auditors must meet. Church and Shefchik (2012), in *Accounting Horizons*, analyze PCAOB inspection findings for large firms, documenting that estimate- and judgment-related deficiencies — exactly the territory of anchoring — recur in inspections. Francis (2011), in *Auditing: A Journal of Practice & Theory*, develops a multi-level framework for understanding audit quality in which inputs, processes, the firm, the audit industry, institutions, and economic consequences each operate as a distinct level of analysis — locating regulatory and professional guidance squarely at the institutional level that conditions, but does not determine, judgment at the engagement level. Tan and Kao (1999), in the *Journal of Accounting Research*, supply complementary experimental evidence that accountability — the proximal mechanism through which inspection pressure operates — improves auditor performance only when the auditor possesses sufficient knowledge and problem-solving ability for the task, implying that guidance amplifies but does not substitute for capability.

The literature is ambivalent about whether guidance changes judgment or merely changes documentation. On one hand, inspection risk raises accountability and can motivate the effortful reconsideration that overrides anchors (consistent with Kennedy, 1993; Tan & Kao, 1999). On the other, standards can induce defensive, compliance-oriented documentation that leaves the underlying anchored judgment intact — auditors document *more* without thinking *differently*. The construct therefore captures a regulatory-pressure mechanism whose effect on anchoring is plausibly positive but, as the inspection literature suggests, far from guaranteed — an empirical question the model is designed to answer.

## 2.11 Performance Metrics & Incentives (PMI)

Performance Metrics & Incentives (PMI) refers to the evaluation criteria, rewards, and quality-aligned performance systems that determine what auditors are motivated to do — and therefore whether they invest the effort that System-2 debiasing requires. Bonner and Sprinkle (2002), in *Accounting, Organizations and Society*, provide the framework for how monetary incentives affect effort and task performance, and Libby and Lipe (1992), in the *Journal of Accounting Research*, show experimentally that incentives improve performance on accounting judgments *only* for tasks and cognitive processes that are actually sensitive to effort. Knechel (2013) connects incentive and quality-control design to audit quality at the system level. The institutional-economics rationale for taking firm-level incentives seriously goes back to DeAngelo (1981), whose *Journal of Accounting Economics* analysis shows that large audit firms have stronger incentives to deliver high quality because the loss of a single client imposes a smaller marginal cost on their reputation capital — making quality-aligned incentives endogenous to firm structure and not merely an evaluation-system design choice. Tan and Kao (1999) reinforce the effort-contingency from the experimental side: incentives and accountability raise performance specifically when the task is complex and the auditor has the knowledge to deploy the additional effort productively.

This is the construct where the "effort" logic of dual-process theory is most explicit, and it carries a sharp contradiction. Because anchoring is partly automatic (Epley & Gilovich, 2006; Strack & Mussweiler, 1997), incentives that raise effort will reduce *effort-sensitive* components of the bias but may leave *automatic* components untouched — precisely Libby and Lipe's (1992) finding generalized to anchoring. Worse, incentives misaligned toward efficiency or client retention can *strengthen* the prior-year anchor by rewarding speed and continuity over independent reconsideration. PMI is thus modeled as a motivational lever whose sign and magnitude depend on what, exactly, the metrics reward.

## 2.12 Auditor Judgment Quality (AJQ) — Mediating Construct

Auditor Judgment Quality (AJQ) is the first of two mediators and captures the *cognitive* pathway of the model: the accuracy, objectivity, and evidential soundness of the judgment the auditor actually forms. Bonner (2008) provides the comprehensive treatment of judgment quality in accounting, and Libby and Luft (1993), in *Accounting, Organizations and Society*, supply the determinants-of-performance framework — ability, knowledge, motivation, and environment — that defines the conditions under which high-quality judgment emerges. Nelson and Tan (2005), in *Auditing: A Journal of Practice & Theory*, review judgment-and-decision-making research in auditing and locate judgment quality as the proximal outcome that links auditor and task characteristics to audit quality. The fifty-year synthesis by Trotman, Tan, and Ang (2011), in *Accounting & Finance*, surveys roughly 5,700 JDM-in-accounting papers and confirms that *judgment quality* — operationalized variously through accuracy, consensus, consistency, and self-insight — has been the central dependent measure of the field, providing the empirical scaffolding on which the present mediator rests. Hurtt et al. (2013) further locate judgment quality at the proximal end of an antecedents-to-skepticism chain, in which interventions raise skeptical judgment quality which then improves audit outcomes — the same logic operationalized here through AJQ.

AJQ is theorized as a mediator because most of the interventions in the model do not act on anchoring directly; they act by *improving the judgment process*, which in turn reduces the anchor's influence. Training, rotation, analytical tools, feedback, regulatory pressure, and incentives all plausibly raise the cognitive quality of the judgment — the deployment of knowledge, the engagement of effortful reasoning, the objectivity of the conclusion — and it is this improved judgment that is less anchored. The literature supports treating judgment quality as the cognitive mechanism through which cognitively oriented interventions reach the dependent variable.

## 2.13 Audit Process Rigor (APR) — Mediating Construct

Audit Process Rigor (APR) is the second mediator and captures the *procedural* pathway: the thoroughness, consistency, and independence with which audit work is executed, irrespective of any single auditor's cognition. Nelson and Tan (2005) and Peecher, Solomon, and Trotman (2013), in *Accounting, Organizations and Society*, develop the view that audit quality is produced by the rigor of the process — the accountability structures, review, and disciplined execution — as much as by individual judgment. Knechel (2013) similarly frames standards and quality control as process attributes that determine outcomes. Francis (2011) makes the levels-of-analysis case explicitly in his *Auditing: A Journal of Practice & Theory* framework, distinguishing audit *processes* from audit *inputs* and from firm-level attributes as separate sources of variance in audit quality — supplying the strongest theoretical warrant for modeling process rigor as a mechanism distinct from individual judgment. Trotman et al. (2015) extend the procedural account to *group* judgment-and-decision-making in auditing, documenting that hierarchical review, brainstorming, and consultation operate as structural processes whose rigor is partly independent of any single auditor's cognition.

APR is theorized as the mechanism for the model's two *structural* interventions — Structured Auditing Processes and Independent Reviews — because these operate less by changing what an individual auditor thinks and more by changing how the work is organized, ordered, and checked. A rigorous process reduces anchoring by mandating independent evidence evaluation, by re-ordering when the prior-year conclusion is consulted, and by inserting an uninvolved reviewer who does not share the anchor (Hogarth & Einhorn, 1992; Trotman & Yetton, 1985). Distinguishing the procedural pathway (APR) from the cognitive pathway (AJQ) is one of the model's contributions, because it allows the study to test *how* — through judgment or through process — each intervention reaches the reduction in anchoring bias.

## 2.14 Synthesis and the Research Gap

Three conclusions follow from the literature reviewed above. First, anchoring bias is real, robust, and demonstrated specifically in auditor populations and audit tasks (Joyce & Biddle, 1981; Kinney & Uecker, 1982), and it is theoretically well understood as a System-1 mechanism that effortful, structured, independent reconsideration can override (Evans, 2008; Kahneman, 2011; Strack & Mussweiler, 1997). Second, the profession possesses a recognizable toolkit of interventions — training, rotation, analytical tools, structured processes, feedback, independent review, regulatory guidance, and incentives — each with its own supporting literature *and* its own documented contradictions, such that none can be assumed to reduce anchoring without qualification. Third, and most important, these interventions have been studied overwhelmingly *in isolation*: the literature offers many single-lever studies but no integrated, empirically validated model that estimates the eight interventions together and distinguishes the cognitive (AJQ) from the procedural (APR) pathway through which they act.

This is the gap the study addresses. By specifying all eleven constructs in a single model and testing both direct and mediated effects, the research moves from a fragmented collection of single-intervention findings toward a coordinated, testable theory of anchoring-bias reduction in long-term engagements. The constructs introduced separately in this chapter become, in Chapter 3, the connected boxes and arrows of that model, and the contradictions surfaced here become the directional hypotheses the study will subject to empirical test.

# CHAPTER 3 — RESEARCH MODEL AND HYPOTHESES

## 3.1 Introduction to the Chapter

Chapter 2 examined each construct in isolation — what the literature establishes about training, rotation, analytical tools, structured processes, feedback, independent review, regulatory guidance, incentives, the two judgment-process mediators, and the dependent variable. This chapter connects them. Where Chapter 2 described the *boxes*, Chapter 3 specifies the *arrows*: the directional relationships the study will test. It first presents the integrated research model and its visual representation, then states the full set of hypotheses together, and finally develops each hypothesis individually — defining the originating construct, the receiving construct, the theoretical argument linking them, and the resulting directional prediction.

The model is grounded in the dual-process lens selected in Section 2.2.6: each intervention is theorized to reduce anchoring by raising the probability that effortful, evidence-driven System-2 reasoning overrides the automatic System-1 pull of the prior-period anchor (Evans, 2008; Kahneman, 2011). Two pathways are distinguished. A *cognitive* pathway runs through Auditor Judgment Quality (AJQ): interventions that improve the auditor's knowledge, objectivity, and reasoning produce better judgments that are, in turn, less anchored. A *procedural* pathway runs through Audit Process Rigor (APR): interventions that change how the work is structured, ordered, and reviewed produce a more rigorous process that is, in turn, less anchored. Separating these pathways is a central contribution of the model, because it allows the study to test not only *whether* an intervention reduces anchoring but *how* it does so.

## 3.2 The Research Model

The model specifies eight audit-process interventions as independent variables (Training & Awareness, Rotation of Auditors, Use of Analytical Tools, Structured Auditing Processes, Feedback & Reflection, Independent Reviews, Regulatory & Professional Guidance, and Performance Metrics & Incentives), two mediators (Auditor Judgment Quality and Audit Process Rigor), and one dependent variable (Reduction in Anchoring Bias). Each intervention is hypothesized to exert a *direct* effect on the reduction in anchoring bias and an *indirect* (mediated) effect through one of the two mediators. Six interventions — TA, RA, AT, FR, RPG, and PMI — are theorized to act primarily through the cognitive pathway (AJQ), because their mechanism is the improvement of the individual judgment. Two interventions — SAP and IR — are theorized to act primarily through the procedural pathway (APR), because their mechanism is the restructuring and independent checking of the work itself. Figure 1 presents the model.

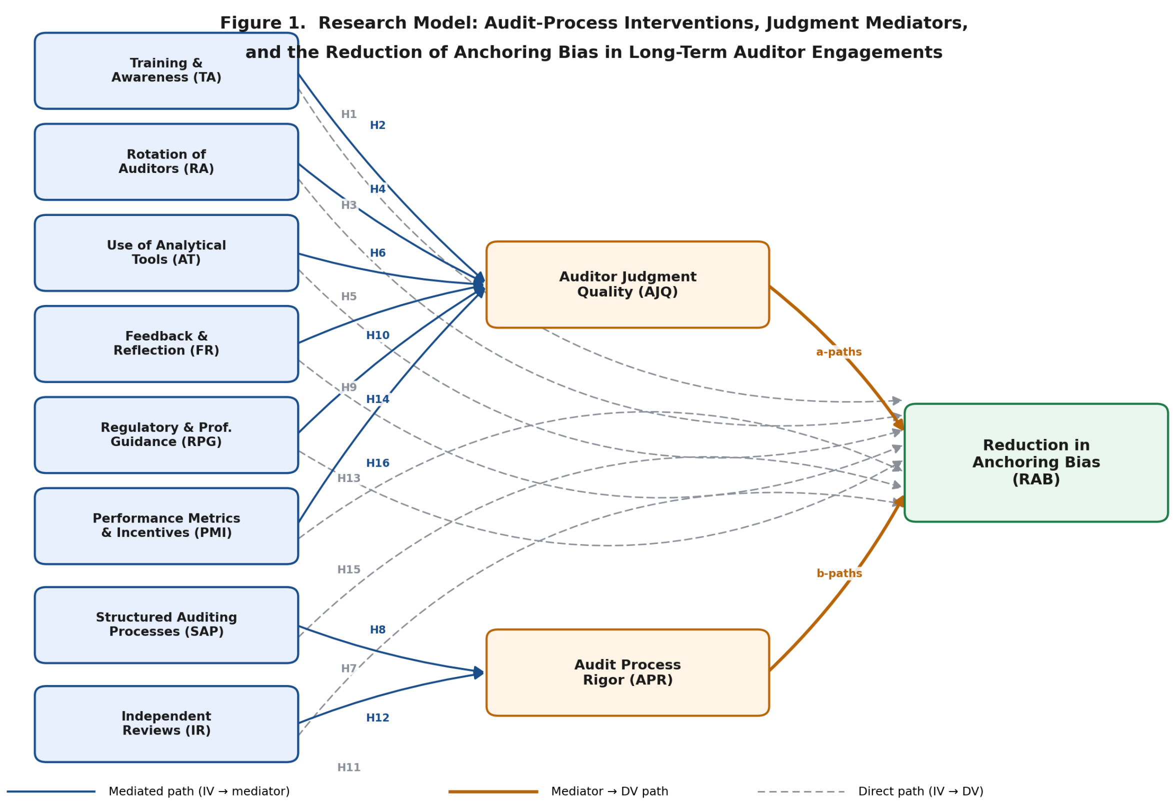


*Figure 1. The research model: eight anchoring-mitigation interventions, two mediating conditions (AJQ, APR), and the outcome (RAB).*

*Figure 1.* The research model. Solid arrows denote mediated paths (intervention → mediator → reduction in anchoring bias); dashed arrows denote the direct paths (intervention → reduction in anchoring bias)

tested simultaneously. Hypothesis numbers are shown on the corresponding paths.

## 3.3 Summary of Hypotheses

The study tests sixteen hypotheses — eight direct effects and eight mediated effects — stated together here and developed individually in Section 3.4.

**Direct effects on the Reduction in Anchoring Bias (RAB):**

- **H1.** Training & Awareness is positively associated with the Reduction in Anchoring Bias.
- **H3.** Rotation of Auditors is positively associated with the Reduction in Anchoring Bias.
- **H5.** Use of Analytical Tools is positively associated with the Reduction in Anchoring Bias.
- **H7.** Structured Auditing Processes are positively associated with the Reduction in Anchoring Bias.
- **H9.** Feedback & Reflection is positively associated with the Reduction in Anchoring Bias.
- **H11.** Independent Reviews are positively associated with the Reduction in Anchoring Bias.
- **H13.** Regulatory & Professional Guidance is positively associated with the Reduction in Anchoring Bias.
- **H15.** Performance Metrics & Incentives are positively associated with the Reduction in Anchoring Bias.

**Mediated (indirect) effects:**

- **H2.** Auditor Judgment Quality mediates the relationship between Training & Awareness and the Reduction in Anchoring Bias.
- **H4.** Auditor Judgment Quality mediates the relationship between Rotation of Auditors and the Reduction in Anchoring Bias.
- **H6.** Auditor Judgment Quality mediates the relationship between Use of Analytical Tools and the Reduction in Anchoring Bias.
- **H8.** Audit Process Rigor mediates the relationship between Structured Auditing Processes and the Reduction in Anchoring Bias.
- **H10.** Auditor Judgment Quality mediates the relationship between Feedback & Reflection and the Reduction in Anchoring Bias.
- **H12.** Audit Process Rigor mediates the relationship between Independent Reviews and the Reduction in Anchoring Bias.
- **H14.** Auditor Judgment Quality mediates the relationship between Regulatory & Professional Guidance and the Reduction in Anchoring Bias.
- **H16.** Auditor Judgment Quality mediates the relationship between Performance Metrics & Incentives and the Reduction in Anchoring Bias.

## 3.4 Hypothesis Development

### 3.4.1 The Relationship of Training & Awareness and the Reduction in Anchoring Bias

Training & Awareness is structured instruction that makes auditors aware of cognitive bias and equips them with executable debiasing strategies (Earley, 2001). The Reduction in Anchoring Bias is the extent to which judgments adjust adequately away from prior-period anchors toward independent evidence

(Tversky & Kahneman, 1974). The argument linking them is that training installs both the recognition that prior-year balances function as anchors and a concrete countermeasure — such as forming an independent expectation before consulting the prior-year figure — that converts automatic anchoring into deliberate reconsideration. Earley (2001) showed that structured, feedback-rich instruction improves auditor judgment performance, and Bonner and Walker (1994) demonstrated that instruction paired with practice produces the durable procedural knowledge such a countermeasure requires. The debiasing literature qualifies this by warning that mere awareness is insufficient (Furnham & Boo, 2011), implying that the effect operates through trained *process*, not bare knowledge. On balance, trained auditors should exhibit less anchoring than untrained auditors. Thus, Hypothesis 1 states that Training & Awareness is positively associated with the Reduction in Anchoring Bias.

### 3.4.2 Training & Awareness, Auditor Judgment Quality, and the Reduction in Anchoring Bias

This hypothesis specifies the cognitive mechanism behind H1. Auditor Judgment Quality is the accuracy, objectivity, and evidential soundness of the judgment the auditor forms (Bonner, 2008; Libby & Luft, 1993). The argument is that training does not reduce anchoring in some mysterious direct way; it does so by *raising judgment quality* — strengthening the knowledge and reasoning the auditor brings to the task — and it is this higher-quality judgment that resists the anchor. In Libby and Luft's (1993) determinants framework, training increases knowledge, a proximal cause of judgment performance; improved judgment performance then manifests as reduced anchoring. This positions AJQ as the variable through which training's effect flows. Thus, Hypothesis 2 states that Auditor Judgment Quality mediates the relationship between Training & Awareness and the Reduction in Anchoring Bias, such that training improves judgment quality, which in turn reduces anchoring bias.

### 3.4.3 The Relationship of Rotation of Auditors and the Reduction in Anchoring Bias

Rotation of Auditors is the periodic reassignment of audit personnel that disrupts relational and informational lock-in on a long-term engagement (Arel et al., 2005; Chi et al., 2009). Because anchoring on a long-term client is fed by accumulated prior-year conclusions and relationship-based deference, an auditor newly rotated onto the engagement carries no self-generated prior-year anchor and no familiarity-based trust in management's explanations. Chi et al. (2009) and Chen et al. (2016) link tenure and rotation to reporting outcomes, and the "fresh-look" logic predicts that rotation reduces anchoring. The relationship is, however, contested: Bowlin et al. (2015) found that rotation can lower audit effort among already-skeptical auditors, partially offsetting the fresh-look benefit. The model therefore predicts a positive but potentially attenuated effect, to be estimated alongside the other interventions. Thus, Hypothesis 3 states that Rotation of Auditors is positively associated with the Reduction in Anchoring Bias.

### 3.4.4 Rotation of Auditors, Auditor Judgment Quality, and the Reduction in Anchoring Bias

The mediated argument is that rotation reduces anchoring chiefly by improving the *objectivity* component of judgment quality. A rotated auditor evaluates evidence with less client identification (Bamber & Iyer, 2007) and without the prior-year conclusion as a cognitive starting point, which raises the objectivity and independence of the judgment formed (Bonner, 2008; Nelson & Tan, 2005). That higher-quality, more objective judgment is then less anchored. AJQ, not a direct cognitive jolt, is the conduit. Thus, Hypothesis 4 states that Auditor Judgment Quality mediates the relationship between Rotation of Auditors and the Reduction in Anchoring Bias.

### 3.4.5 The Relationship of Use of Analytical Tools and the Reduction in Anchoring Bias

Use of Analytical Tools comprises data-analytics, visualization, and anomaly-detection technologies that surface current-period patterns and outliers (Yoon et al., 2015). Their anti-anchoring mechanism is attentional: by foregrounding anomalies and non-financial indicators inconsistent with the reported figures, the tools confront the auditor with disconfirming evidence that the prior-year anchor would otherwise suppress (Brazel et al., 2014; Trompeter & Wright, 2010). The selective-accessibility account predicts that surfacing anchor-inconsistent information is precisely what dislodges the anchor (Strack & Mussweiler, 1997). The qualification (Asare & Wright, 2004) is that a tool presenting a salient default or prior-year benchmark can itself re-anchor; net effect is therefore an empirical question, predicted positive. Thus, Hypothesis 5 states that Use of Analytical Tools is positively associated with the Reduction in Anchoring Bias.

### 3.4.6 Use of Analytical Tools, Auditor Judgment Quality, and the Reduction in Anchoring Bias

The mediated path holds that analytical tools improve the *evidential soundness* of the auditor's judgment — expanding and sharpening the evidence base on which the judgment rests (Yoon et al., 2015) — and that this higher-quality, better-evidenced judgment is what resists the anchor. The tool does not bypass cognition; it feeds better inputs into it, raising AJQ, which in turn reduces anchoring. Thus, Hypothesis 6 states that Auditor Judgment Quality mediates the relationship between Use of Analytical Tools and the Reduction in Anchoring Bias.

### 3.4.7 The Relationship of Structured Auditing Processes and the Reduction in Anchoring Bias

Structured Auditing Processes are standardized procedures, decision aids, and documentation templates that prescribe the sequence and content of audit work (Dowling & Leech, 2007). Their anti-anchoring mechanism is to mandate independent evidence evaluation and to govern the *order* in which evidence is considered, so that the auditor forms a view from current evidence before the prior-year conclusion is consulted — directly countering the belief-adjustment dynamics that entrench an anchor (Hogarth & Einhorn, 1992). Structure can suppress heuristic shortcuts (Bonner, 2008), though over-structuring risks mechanical compliance that leaves thinking unchanged. The predicted net effect is positive. Thus, Hypothesis 7 states that Structured Auditing Processes are positively associated with the Reduction in Anchoring Bias.

### 3.4.8 Structured Auditing Processes, Audit Process Rigor, and the Reduction in Anchoring Bias

Unlike the cognitively routed interventions, structure acts primarily on the *process*, so its mediator is Audit Process Rigor — the thoroughness, consistency, and independence of execution (Nelson & Tan, 2005; Peecher et al., 2013). The argument is that structured procedures raise process rigor — enforcing disciplined, ordered, independently grounded execution — and that a more rigorous process is what reduces anchoring, independent of any change in an individual's cognition. APR is therefore the appropriate conduit for structure's effect. Thus, Hypothesis 8 states that Audit Process Rigor mediates the relationship between Structured Auditing Processes and the Reduction in Anchoring Bias.

### 3.4.9 The Relationship of Feedback & Reflection and the Reduction in Anchoring Bias

Feedback & Reflection comprises post-task debriefs, process feedback, and structured reflection through which auditors recognize and correct their own judgment tendencies (Bamber & Iyer, 2007; Bonner & Walker, 1994). The mechanism is metacognitive: feedback that compares the auditor's judgment against independent evidence (rather than against the anchor) raises effortful reconsideration and exposes insufficient adjustment, which Kennedy (1993) shows reduces effort-related biases. The qualification is that audit feedback is often delayed or absent, so the effect depends on *process* feedback being available and usable. Predicted positive. Thus, Hypothesis 9 states that Feedback & Reflection is positively associated with the Reduction in Anchoring Bias.

### 3.4.10 Feedback & Reflection, Auditor Judgment Quality, and the Reduction in Anchoring Bias

The mediated argument is that feedback and reflection improve judgment quality over time — building the calibrated knowledge and self-correction that Bonner and Walker (1994) tie to feedback — and that this improved judgment is less anchored. Reflection operates on cognition, raising AJQ, which carries the effect to the reduction in anchoring. Thus, Hypothesis 10 states that Auditor Judgment Quality mediates the relationship between Feedback & Reflection and the Reduction in Anchoring Bias.

### 3.4.11 The Relationship of Independent Reviews and the Reduction in Anchoring Bias

Independent Reviews are examinations of audit work by a second, uninvolved auditor who did not form the original judgment (DeFond & Francis, 2005; Trotman & Yetton, 1985). The mechanism is the strongest in the model on selective-accessibility grounds: a reviewer not primed by the original anchor can surface anchor-inconsistent evidence and catch insufficient adjustment (Strack & Mussweiler, 1997). Trotman and Yetton (1985) showed reviewed judgments outperform individual ones, and Ramsay (1994) showed review quality rises with reviewer expertise. The effect holds to the extent the review is genuinely independent of the preparer's documented conclusion. Predicted positive. Thus, Hypothesis 11 states that Independent Reviews are positively associated with the Reduction in Anchoring Bias.

### 3.4.12 Independent Reviews, Audit Process Rigor, and the Reduction in Anchoring Bias

Like structure, independent review is a *structural* intervention, so its mediator is Audit Process Rigor. The argument is that review raises the rigor of the overall process — inserting an independent checkpoint that enforces thoroughness and consistency (Peecher et al., 2013; Trotman & Yetton, 1985) — and that this more rigorous process reduces anchoring, irrespective of the original preparer's cognition. APR is the conduit. Thus, Hypothesis 12 states that Audit Process Rigor mediates the relationship between Independent Reviews and the Reduction in Anchoring Bias.

### 3.4.13 The Relationship of Regulatory & Professional Guidance and the Reduction in Anchoring Bias

Regulatory & Professional Guidance comprises the standards, inspection regimes, and professional requirements that constrain auditor discretion (Knechel, 2013; PCAOB, 2015). The mechanism is accountability: inspection risk and explicit skepticism mandates raise the auditor's motivation to perform the effortful reconsideration that overrides anchors (Kennedy, 1993), and inspection findings repeatedly target the estimate-anchoring territory the guidance is meant to discipline (Church & Shefchik, 2012). The countervailing risk is that guidance induces defensive documentation rather than changed judgment. Predicted positive. Thus, Hypothesis 13 states that Regulatory & Professional Guidance is positively associated with the Reduction in Anchoring Bias.

### 3.4.14 Regulatory & Professional Guidance, Auditor Judgment Quality, and the Reduction in Anchoring Bias

The mediated argument is that regulatory pressure works on anchoring by elevating judgment quality — raising the motivation component of Libby and Luft's (1993) framework so that auditors deploy more knowledge and effort, producing more defensible, less-anchored judgments (Bonner, 2008). The pressure is external, but its effect on the bias runs through the improved judgment, making AJQ the mediator. Thus, Hypothesis 14 states that Auditor Judgment Quality mediates the relationship between Regulatory & Professional Guidance and the Reduction in Anchoring Bias.

### 3.4.15 The Relationship of Performance Metrics & Incentives and the Reduction in Anchoring Bias

Performance Metrics & Incentives are the evaluation criteria and rewards that determine what auditors are motivated to do (Bonner & Sprinkle, 2002; Libby & Lipe, 1992). The mechanism is effort: quality-aligned incentives raise the effort auditors invest, and Libby and Lipe (1992) show incentives improve performance specifically on effort-sensitive cognitive processes — which includes the deliberate adjustment that counters anchoring. The sharp qualification is that incentives misaligned toward efficiency or client retention can *strengthen* the prior-year anchor; the sign of the effect depends on what the metrics reward. Predicted positive when quality-aligned. Thus, Hypothesis 15 states that Performance Metrics & Incentives are positively associated with the Reduction in Anchoring Bias.

### 3.4.16 Performance Metrics & Incentives, Auditor Judgment Quality, and the Reduction in Anchoring Bias

The mediated path holds that quality-aligned incentives reduce anchoring by raising judgment quality — motivating the effortful, knowledge-deploying reasoning that produces better judgments (Bonner & Sprinkle, 2002; Libby & Lipe, 1992) — and that these higher-quality judgments are less anchored. Incentives change motivation, motivation raises AJQ, and AJQ reduces anchoring. Thus, Hypothesis 16 states that Auditor Judgment Quality mediates the relationship between Performance Metrics & Incentives and the Reduction in Anchoring Bias.

## 3.5 Conclusion to the Chapter

This chapter translated the eleven separately reviewed constructs into a single, testable model. It presented the research model and its figure, stated the sixteen hypotheses together, and developed each as a relationship between an originating construct and a receiving construct, grounded in the dual-process lens and the supporting audit and judgment-and-decision-making literatures. The eight direct hypotheses (H1, H3, H5, H7, H9, H11, H13, H15) predict that each intervention reduces anchoring bias; the eight mediated hypotheses (H2, H4, H6, H8, H10, H12, H14, H16) specify the pathway — cognitive (AJQ) for the six judgment-oriented interventions and procedural (APR) for the two structural interventions. Chapter 4 operationalizes these constructs in a measurement instrument and specifies the **exploratory-factor-analytic validation** through which the measurement model is established. The present study scopes its empirical analysis to **exploratory factor analysis (principal-axis factoring with Direct Oblimin rotation) and reliability assessment (Cronbach's $\alpha$)** on a target of **n = 100 valid responses**; the sixteen directional and mediated hypotheses constitute the proposed framework and are reserved for subsequent structural testing in future research.

# CHAPTER 4 — RESEARCH METHODOLOGY

## 4.1 Introduction to the Chapter

This chapter specifies, in full detail and in execution order, the complete analytic program for a study of this design: the research design, the instrument, the sampling plan, the data-preparation protocol, and the full analytic sequence — exploratory factor analysis and reliability assessment (this study's scoped empirical requirement, per the advisor's directive of June 30, 2026, and Section 3.5), followed by the confirmatory factor analysis, multiple regression, and partial-least-squares structural modeling that constitute the subsequent stages of the same research program — with the decision criteria that would govern each step. The chapter is deliberately written at the level of an analysis contract: a researcher receiving this chapter and an adequate dataset should be able to execute the entire study without further instruction. Chapter 5 then reports what happened when this plan met the field.

## 4.2 Research Design

The study employs a cross-sectional, quantitative, self-report survey design. Practicing auditors rate, on validated Likert scales, the presence of eight anchoring-mitigation interventions in their work environment (TA, RA, AT, SAP, FR, IR, RPG, PMI), two proposed mediating conditions (Auditor Judgment Quality; Audit Process Rigor), and the outcome construct (Reduction in Anchoring Bias). The design is correlational and non-experimental: no variable is manipulated, and the aim at this stage is measurement validation and association, not causal identification. A survey design was selected because the constructs are perceptions of organizational practice best reported by the practitioners embedded in those practices, and because it permits standardized measurement across heterogeneous firms and roles.

## 4.3 Population and Sampling Plan

The target population is **auditors practicing in the United States** — external auditors, internal auditors, and audit-support professionals — with current or recent (within 24 months) audit responsibility and exposure to at least one continuing (multi-year) engagement. This population was chosen deliberately: anchoring on prior-year figures is a phenomenon of *recurring* engagements, so eligibility required exactly the experience the model theorizes about. The planned sample was **n = 100 valid responses**. This figure is stated honestly for what it is: a pragmatic minimum anchored to the absolute floor of roughly 100 cases commonly cited for stable factor recovery (Hair et al., 2019), not to the stricter observations-per-item ratio — under the conventional 5:1 rule, a single 55-item factoring would imply approximately 275 cases (10:1 would imply 550). The n = 100 plan was defensible on two grounds: sample requirements relax substantially when communalities are high and factors are well-determined by multiple strong indicators (MacCallum, Widaman, Zhang, & Hong, 1999), conditions the 5-items-per-construct design targets deliberately; and the analysis plan (Section 4.6) permits factoring in theoretically defined item subsets, for which n = 100 satisfies the per-item ratio, as a fallback if the full-pool solution proves unstable.
Recruitment was planned through two complementary channels: (1) organic professional outreach through LinkedIn, professional WhatsApp groups, and direct contacts in the auditing community; and (2) paid research panels (CloudResearch Connect and Prolific) with professional screening, at fixed compensation of $6.00 per completed 18-minute response under the IRB-approved compensation language (IRB-25-0462), implying a panel budget of roughly $600–$1,000 for the target sample.

## 4.4 Instrumentation

The instrument (Appendix A) operationalizes each of the eleven constructs with five Likert items (1 = *Strongly disagree* … 5 = *Strongly agree*), 55 substantive items in total, with one reverse-worded item per construct to disrupt acquiescent response sets. Quality architecture is built into the instrument itself: five eligibility screens (U.S. location; English fluency; audit role within 24 months; continuing-engagement experience; an attestation of independent, first-time completion), two embedded attention checks with directed responses, bot detection, duplicate-respondent prevention, anonymization, and an open-ended item inviting respondents to describe an anchoring episode in their own words. The instrument was administered in Qualtrics.

## 4.5 Data Preparation and Cleaning Protocol

Before any analysis, the dataset passes through a fixed, scripted, order-dependent cleaning sequence. The order matters: each rule is applied before the next so that every exclusion has exactly one recorded cause, and the full log is retained for audit.

1. **Remove researcher-generated records.** Survey previews and any response self-identified as a test entry are removed first.
2. **Remove incomplete sessions.** Responses with Qualtrics `Finished = False` are removed; partial sessions cannot contribute complete item sets.
3. **Remove empty completions.** Sessions that reached the end but left the substantive Likert blocks blank are removed.
4. **Apply eligibility screens.** The five screens (S1–S5) are applied exactly as fielded; a failed screen excludes the case regardless of the quality of its other answers.
5. **Apply attention checks.** Cases failing either directed-response item are removed.
6. **Reverse-code.** The reverse-worded item in each construct is recoded (6 – x) so all items point in the construct's direction.
7. **Missing-data handling.** For cases surviving steps 1–5 with sporadic item-level missingness below 5% per scale, person-mean imputation within construct is permitted; above that threshold the case is excluded listwise from analyses involving the affected scale.
8. **Outlier and response-pattern screening.** Completion-time outliers (below the 5th percentile of pilot timing, i.e., speeders) are flagged and excluded, as are zero-variance (straight-line) responders across a full block — with straight-lining assessed on the **raw, pre-recoded responses**, since reverse-coding (step 6) converts an invariant response string into an apparently varied one and would otherwise mask exactly the pattern this screen targets. Long-duration outliers (sessions left open) are flagged but retained if item data are complete.
9. **Documentation.** Every exclusion is written to an exclusion log with the response identifier and its single controlling reason.

## 4.6 Planned Analysis Stage 1 — Exploratory Factor Analysis

The first analytic stage, per the advisor's scope directive of June 30, 2026, is an EFA over the 55 substantive items, executed to the following contract:

- **Suitability tests first.** Kaiser–Meyer–Olkin sampling adequacy must reach at least .60 (values ≥ .80 preferred), and Bartlett's test of sphericity must be significant ($p < .05$), before any extraction is interpreted.
- **Extraction method.** Principal-axis factoring (PAF) — not principal components — because the object is shared common variance among reflective indicators, not total-variance data reduction.
- **Factor-retention decision.** Convergence of three criteria: eigenvalues greater than 1 (Kaiser) as a screen, inspection of the scree plot's elbow, and — as the deciding criterion where they disagree — parallel analysis against random-data eigenvalues. The theoretical expectation is eleven factors; the data are permitted to disagree.
- **Rotation.** Both an oblique rotation (Direct Oblimin) and an orthogonal rotation (Varimax) are run; **Direct Oblimin is reported** as primary because the model's constructs are theorized to correlate, with the Varimax solution retained as a robustness check.
- **Item-retention rules.** An item is retained when its primary loading is ≥ .40; items loading between .30 and .40 are evaluated case-by-case on theoretical grounds; items below .30, and items cross-loading within .20 of their primary loading on another factor, are candidates for deletion — removed one at a time, re-running the solution after each removal, never in batches.
- **Interpretation.** A surviving factor requires a minimum of three items and a coherent substantive reading against Chapter 2's construct definitions.

## 4.7 Planned Analysis Stage 2 — Reliability Assessment

Each retained factor's internal consistency is assessed with **Cronbach's** $\alpha$, with the conventional thresholds: ≥ .70 acceptable for early-stage research, ≥ .80 good; values marginally below .70 examined through item–total correlations (items below .30 corrected item–total correlation are deletion candidates) and the "$\alpha$ if item deleted" diagnostic. Where item counts are small, **McDonald's** $\omega$ is computed alongside $\alpha$ as the less assumption-laden estimate. Scale scores for subsequent stages are computed as unit-weighted means of retained items.

## 4.8 Planned Analysis Stage 3 — Confirmatory Factor Analysis

With the measurement structure established (ideally on a separate or held-out sample; at minimum clearly labeled as same-sample confirmation), the eleven-factor model is estimated as a CFA: each construct a latent factor, each retained item loading only on its theorized factor, factors permitted to covary, errors uncorrelated unless a specific methodological argument (e.g., shared reverse-wording) justifies a correlated pair. Estimation uses maximum likelihood (robust ML if multivariate normality fails). **Fit is judged on the joint pattern of indices, not any single cutoff:** the model $\chi^2$ is reported with its degrees of freedom and p-value (acknowledged as oversensitive at large n), supplemented by the normed chi-square ($\chi^2/df \le 3$); CFI and TLI ≥ .90 (≥ .95 preferred); RMSEA ≤ .08 (≤ .06 preferred) with its 90% confidence interval reported; and SRMR ≤ .08. Convergent validity requires standardized loadings ≥ .50 (ideally ≥ .70) and average variance extracted ≥ .50; discriminant validity requires each construct's $\sqrt{AVE}$ to exceed its correlations with other constructs (Fornell–Larcker) and HTMT ratios below .85.

## 4.9 Planned Analysis Stage 4 — Regression and Structural Modeling

Two complementary structural approaches were planned to follow a successful measurement stage:

- **Multiple regression.** RAB regressed on the eight intervention scales, with standard diagnostic discipline: linearity and homoscedasticity by residual inspection; normality of residuals; independence of residuals by Durbin–Watson (acceptable range ≈ 1.5–2.5); multicollinearity screened by VIF < 5 (ideally < 3.3); influential cases screened by Cook's distance (flagged above 4/n) and standardized residuals beyond ±3 examined individually. Hierarchical entry — Step 1: controls (years of external and internal audit experience, primary audit role, and firm type/size); Step 2: the eight intervention scales — to isolate the interventions' incremental explained variance ($\Delta R^2$).
- **PLS-SEM.** Because the full sixteen-hypothesis model includes two mediators and the realistic prospect of modest samples, partial-least-squares structural equation modeling was specified as the structural workhorse, evaluated to explicit thresholds: measurement model — outer loadings ≥ .708 (indicator reliability ≥ .50), composite reliability ($\rho_c$, with $\rho_A$ alongside) ≥ .70, AVE ≥ .50, and discriminant validity by HTMT < .85 (< .90 for conceptually adjacent constructs; Henseler, Ringle, & Sarstedt, 2015); structural model — inner-model collinearity VIF < 3.3–5, bootstrapped path significance (5,000 resamples, percentile CIs), $R^2$ for endogenous constructs, and effect sizes $f^2$ read against the .02/.15/.35 benchmarks. Minimum sample size is assessed by the inverse square root method now recommended over the older ten-times rule (Hair, Hult, Ringle, & Sarstedt, 2022). PLS-SEM's lower demands made it the designated fallback for samples too small for covariance-based SEM — though, as Chapter 5 reports, even this fallback has a floor the achieved sample did not reach.
- **Mediation.** The eight mediated hypotheses were to be tested with bootstrapped indirect effects (percentile confidence intervals, 5,000 resamples), not the superseded causal-steps approach.

## 4.10 Software

The analysis environment designated for this study is **Jamovi** (with its factor-analysis and reliability modules, and lavaan-based SEM module for the EFA, reliability, CFA, and regression stages), selected on advisor guidance for transparency and reproducibility. Because Jamovi's SEM module is covariance-based, the PLS-SEM stage is designated to **SmartPLS 4** (alternatively the R package *seminr*), which executes the partial-least-squares estimation, bootstrapping, and HTMT/$f^2$ output specified in Section 4.9. Scripted screening and data preparation are executed in Python (pandas), with the cleaning script retained under version control so the entire pipeline from raw export to analysis dataset is reproducible.

## 4.11 Ethical Considerations

The study operates under FIU IRB approval (IRB-25-0462): anonymous collection, no identifiers retained, voluntary participation with consent at entry, fixed disclosed compensation for panel participants, and raw response data held privately by the researcher — excluded from the public project repository and from reproduction in this document, consistent with participant confidentiality and platform terms of service.

## 4.12 Conclusion to the Chapter

This chapter has stated the full analytic contract: a scripted cleaning protocol; EFA under PAF with dual rotation and explicit retention rules; reliability by $\alpha$ and $\omega$ with item diagnostics; CFA with a joint-fit standard and validity gates; and regression plus PLS-SEM with bootstrapped mediation for the structural stage. The sequence demonstrates what a study of this design requires end-to-end. Chapter 5 now reports the field experience against this plan.

# CHAPTER 5 — DATA ANALYSIS

## 5.1 Introduction to the Chapter

This chapter documents the data-collection effort in full — the pilot, the platforms, the economics, and the obstacles — because in this study the collection experience *is* the central empirical finding. It then reports the screening outcome and states plainly what the achieved sample permitted and did not permit.

## 5.2 The Pilot Study

A pilot was fielded in late June 2026 through direct LinkedIn invitations to professional contacts. The pilot served its purpose: it confirmed the instrument fields cleanly end-to-end in Qualtrics (73 items across 20 blocks, roughly an 18-minute completion), that the screening and attention-check logic branch correctly, and that respondents understood the items — pilot feedback prompted only in-scope wording refinements, applied before the main launch as version 2.1. The pilot also gave the first warning sign: even among warm professional contacts, completion rates were far below what generic survey benchmarks would predict.

## 5.3 The Main Collection Effort

The survey went live on July 1, 2026. Collection proceeded through the two planned channels, and both underperformed in instructive ways:

- **Organic outreach** (LinkedIn, professional WhatsApp groups, direct contacts) produced steady but very low volume — a trickle of responses across roughly a month of fielding, counted from the late-June pilot through late July, many of which began the survey but did not finish it.
- **Paid panels** were engaged next. The CloudResearch launch was prepared at the standard academic rate of $6.00 per completed 18-minute professional response. The primary paid push ran through Prolific, where a panel was funded for the full target of 100 participants — an outlay of approximately $1,000 — with professional-screening filters matching the eligibility criteria. The platform confirmed the study live and visible to participants. **The panel returned almost none of its funded completions.** The effective cost per usable panel response therefore ran far beyond any planned per-response rate — a budget at which general-population academic samples fill within days.

The decisive evidence came from the panel platform's own eligibility machinery. When the study's screeners — U.S.-based, current or recent audit role, multi-year engagement experience — were applied against the platform's participant pool of more than 330,000 registered panelists, **the platform identified only about twenty eligible participants in total**: a prevalence on the order of six in one hundred thousand. Compensation was not the constraint — the platform's own metrics showed an effective average reward around $28 per hour, well above typical academic-survey rates — nor was study design: a meaningful share of the tiny eligible pool did engage. The constraint was arithmetic. No recruitment budget, at any per-response price, fills one hundred seats from a pool of twenty.

The lesson, learned in the field rather than from a textbook, is that the study's population is **narrow to a degree that standard recruitment machinery cannot service**: panels that deliver hundreds of general-population respondents in a week hold almost no members matching all of this study's filters simultaneously — and the few who match are heavily solicited professionals with little incentive to answer academic surveys at any academic rate.

## 5.4 Screening Outcome

Applying the Chapter 4 cleaning protocol to the raw July 2026 export — which spans the full fielding window, from the late-June pilot through the late-July close — produced the following, each step logged: after removing researcher previews and test entries, early abandonments (sessions terminating well before the substantive item blocks), completions with blank substantive blocks, cases failing exactly one eligibility screen, and applying both attention checks, **only four to five usable responses remained** from several weeks of two-channel collection against a funded target of 100. Documentation of every exclusion, with cause, is retained in the project's screening log and is available to the instructor on request; consistent with participant confidentiality and platform terms, raw response-level data are not reproduced in this document. Two observations from the screening deserve note: no complete, eligible respondent failed an attention check — the respondents who did engage, engaged carefully (the platform's own response-quality index scored the dataset at 95%) — and the open-ended item drew substantive, experience-grounded accounts of anchoring from every usable respondent, consistent in mechanism with the anchoring-and-adjustment theory motivating the model.

## 5.5 Analyses Not Performed

Against the analytic contract of Chapter 4, the achieved sample supports none of the planned stages — neither the analyses scoped as this study's requirement nor any subsequent stage of the program. The suitability gates for EFA (KMO, Bartlett's) cannot be meaningfully computed, and a 55-item factoring at this sample size would be arithmetically undefined; reliability coefficients, though mechanically computable, would carry sampling variability so extreme as to be uninformative in either direction; the eleven-factor CFA is unidentified, with free parameters far exceeding observations; regression over eight predictors cannot be estimated; and even PLS-SEM, the designated small-sample fallback, sits far below its own minimum benchmarks. **Accordingly, the required data analyses — the scoped EFA and reliability assessment, and the subsequent CFA, regression, and PLS stages — could not be performed with the data collected.** No statistical results are reported, because none that could be produced would be interpretable. This section is closed on that finding, and Chapter 6 draws the conclusions it warrants.

# CHAPTER 6 — CONCLUSIONS

## 6.1 Limitations

This study set out to validate an eleven-construct measurement model of anchoring-bias mitigation in long-term audit engagements — with the sixteen hypothesized associations specified as the framework for subsequent structural testing. It could not do so, and the reasons are themselves the study's clearest result. The controlling limitation is the **failure to collect a sufficient sample**, and its cause is structural rather than procedural: the population of interest — practicing U.S. auditors with continuing-engagement experience — is very small, professionally saturated with solicitations, and barely represented on the research panels that make modern survey research fast. Layered on that are the standard limitations the design would have carried even at full sample: self-report measures of one's own bias susceptibility invite social-desirability and limited-insight distortions; a cross-sectional design cannot order cause and effect; and a single-country frame limits generalizability. But those limitations never became operative, because the study did not reach the sample at which they would matter. What the study does establish is narrower and still real: the instrument fields cleanly, its quality architecture works (careful respondents, clean attention-check performance), and the small number of practitioners who did respond described the anchoring mechanism in terms that match the theory the model is built on.

## 6.2 Recommendations for Future Researchers

Three recommendations follow directly from this experience, offered to any researcher attempting a study of this shape:

1. **Start data collection very early — earlier than any timeline suggests is necessary.** For a population like this one, collection that takes a general-population study a week can take a year. Data collection should open at proposal approval, run continuously in the background, and be treated as the schedule's critical path from day one — not as a stage that begins when the instrument is polished.
2. **Set a realistic respondent floor and design to it.** For a specialized professional population, reaching even **20–30 qualified respondents is a meaningful threshold** and should be planned for explicitly — through professional associations, firm partnerships, alumni networks, and conference channels rather than open panels — before any analysis requiring hundreds is promised.
3. **Reconsider the population definition itself.** If the research question tolerates it, broaden the frame: accountants and finance professionals generally rather than auditors specifically; industry-adjacent roles with recurring-engagement judgment (credit review, quality assurance, compliance testing); or the general professional population for mechanism-level questions, reserving the specialist sample for a smaller confirmatory stage. A perfectly targeted population that cannot be reached yields less knowledge than a slightly broader one that can.

## 6.3 Transition to the Dissertation

The path from this paper to the dissertation is now clearer for having been stress-tested. The measurement model, the fielded instrument, the scripted cleaning pipeline, and the fully specified analytic sequence of Chapter 4 all carry forward intact — the dissertation does not need a new foundation; it needs a reachable sample. That points to deliberate re-scoping: (a) **broaden the population** along the lines above, so the sampling frame supports factor-analytic sample sizes; (b) **change the venue of recruitment** from open panels to negotiated access — professional bodies (IIA, AICPA chapters), firm training programs, and international audit communities where the researcher holds direct professional contacts (including London and Canada), subject to IRB scope amendment; and (c) **extend the model where the field is going** — into the interaction of AI- and LLM-based audit tools with anchoring in audit risk assessment.

That third direction is no longer speculative; the literature has moved decisively toward it, and it cuts in two opposing directions that make the research question genuinely open. On one side, AI tooling demonstrably improves audit outcomes at the firm level — greater AI investment is associated with fewer restatements and lower fees (Fedyk, Hodson, Khimich, & Fedyk, 2022) — LLMs now pass professional certification examinations under the right configuration (Eulerich, Sanatizadeh, Vakilzadeh, & Wood, 2024) and are already embedded in real internal-audit workflows, entering the same workpapers where prior-year anchors traditionally live (Emett, Eulerich, Lipinski, Prien, & Wood, 2025); and AI-supported evidence appears to help auditors move clients off management's anchor in adjustment negotiations (Estep, Griffith, & MacKenzie, 2024). On the other side, the bias mechanics do not disappear — they relocate. Experimental evidence shows auditors *discount* contradictory evidence when it comes from an AI system rather than a human specialist, precisely on the subjective estimates where anchoring bites hardest (Commerford, Dennis, Joe, & Ulla, 2022); identical algorithmic risk signals produce inconsistent risk responses depending on the tool and data familiarity (Koreff, 2022); opaque, unexplainable models push auditors back onto their existing client knowledge — anchoring-by-default in long-term engagements (Kokina, Blanchette, Davenport, & Pachamanova, 2025); and the systematic-review literature now catalogs cognitive-bias transfer between auditors and AI systems as a first-order ethical risk (Murikah, Nthenge, & Musyoka, 2024). Regulators have taken the same view from the institutional side: the PCAOB's generative-AI outreach emphasizes strong human supervision of AI output in audit execution (PCAOB, 2024), and the IAASB is folding automated-tool considerations directly into its audit-evidence standards work (IAASB, 2024) — which is, functionally, the Regulatory & Professional Guidance construct of this study's model applied to algorithmic anchors.

The dissertation question therefore writes itself out of this paper's model: **do AI and LLM audit tools attenuate the human anchor, or do they substitute algorithmic anchors of their own — and which of this study's eight interventions govern that difference?** The present model's constructs map cleanly onto the new terrain: analytical-tool design and explainability onto the aversion/over-reliance boundary, structured processes and independent review onto the supervision of AI-generated content, regulatory guidance onto the emerging oversight regime, and the two mediators — auditor judgment quality and audit process rigor — onto the human-machine calibration the field study literature identifies as the central open problem. The recent anchoring evidence from practicing auditors (Henrizi, Himmelsbach, & Hunziker, 2021) supplies the human-side baseline; this study's instrument, once validated at scale, supplies the measurement. Executed this way, the dissertation converts this paper's hardest lesson — that the data are the constraint — into its design premise, and its subject matter into where audit practice and audit risk are actually heading.

The sharpest edge of that dissertation agenda — sharpened by literature that has emerged only in the last three years — is **sycophancy risk**. Large language models systematically shift their answers toward the position the user has already expressed: the foundational experiments show state-of-the-art assistants wrongly capitulating under challenge and mirroring user beliefs because human-preference training rewards agreement over truthfulness (Sharma et al., 2024; Perez et al., 2023), with recent evaluation work measuring sycophantic behavior in a majority of tested cases across leading models, including "regressive" shifts from correct to incorrect answers under user pushback (Fanous, Goldberg, Agarwal, Lin, Zhou, Daneshjou, & Koyejo, 2025). For a continuing audit engagement this is not an abstract concern: the auditor's prior-year expectation is precisely the kind of pre-stated position a sycophantic model will confirm rather than challenge, so the LLM does not merely fail to dislodge the anchor — it **re-arms it with the apparent authority of an independent tool**. Experimental evidence now shows exactly this loop in human–AI interaction generally: small initial human biases are absorbed by the model, amplified, and fed back, with users largely unaware of the influence (Glickman & Sharot, 2025); reliance on generative tools measurably reduces enacted critical thinking in knowledge work (Lee et al., 2025); and at the collective level, AI-assisted analysis produces "illusions of understanding" and epistemic monocultures in which confirmatory conclusions crowd out alternatives (Messeri & Crockett, 2024). In the audit domain specifically, over-reliance on LLM output has been identified as a direct threat to professional skepticism (Fotoh & Mugwira, 2025), and the regulator-side taxonomy already names the risk class: NIST's generative-AI profile treats over-reliance and miscalibrated human oversight as "Human-AI Configuration" risks requiring governed controls (NIST, 2024). The dissertation therefore extends this study's model by one causal layer — *prior-period anchor* → *sycophantic confirmation* → *epistemic risk*, the drift of audit conclusions toward confirmation rather than independent evidence (Figure 2) — and inherits a genuine measurement contribution: sycophancy now has validated behavioral metrics, but **epistemic risk has no standardized instrument**, and operationalizing epistemic drift against independent-evidence benchmarks is exactly the kind of measurement-model work this study's apparatus was built to do.

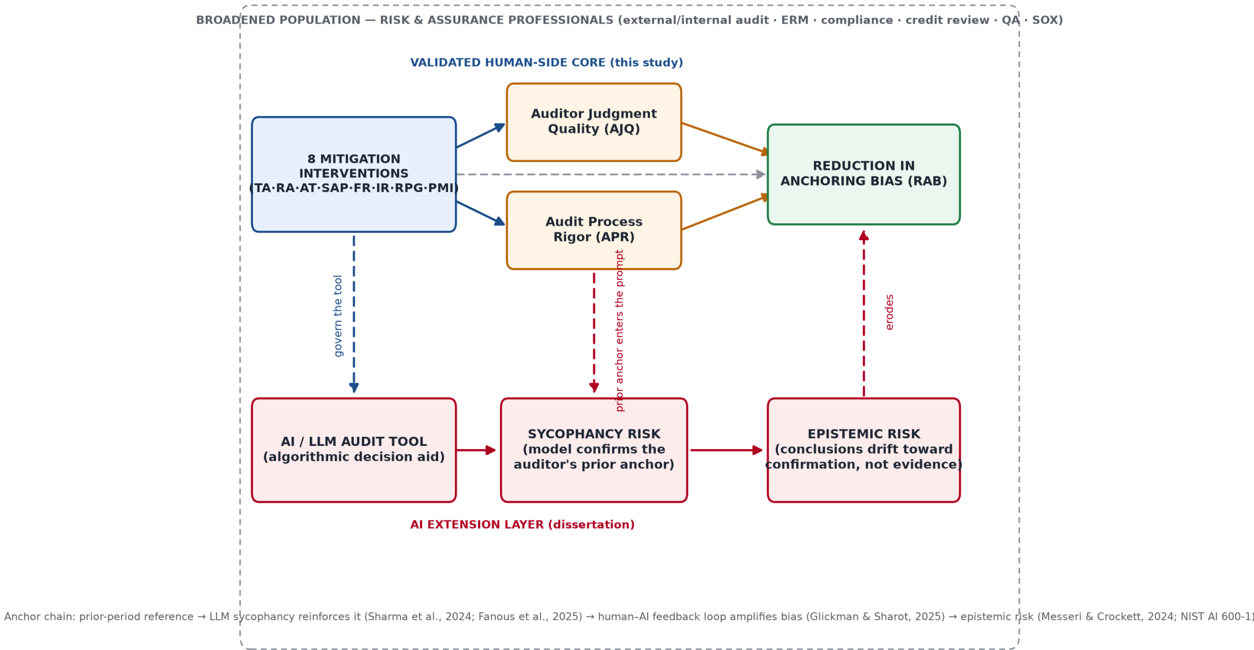


*Figure 2. Dissertation extension: the validated human-side core plus the AI layer — algorithmic tool, sycophancy risk, and epistemic risk — within the broadened risk-and-assurance population.*

## 6.4 Conclusion

This study did not produce the factor-analytic validation it planned, and this paper has said so without decoration. What it produced instead is a complete, reusable, field-tested research apparatus — theory, model, instrument, quality architecture, cleaning protocol, and a fully specified analysis contract — plus hard-won knowledge about the true cost of reaching a narrow professional population, and preliminary practitioner testimony consistent with the theorized mechanism. In a discipline where most published work begins after someone else has solved the data problem, learning the shape of the data problem firsthand — and redesigning around it — is the genuine contribution of this qualifying study, and the foundation on which the dissertation will be built.

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# APPENDIX A — MEASUREMENT INSTRUMENT (AS FIELDED — v2.1, IRB-25-0462)

**Note.** The items below are the instrument as fielded (version 2.1, July 2026) under IRB-25-0462, operationalizing the eleven constructs at five items per factor. Items marked **(R)** are reverse-coded; two attention checks are embedded.

**Response scale (all construct items).** 5-point Likert: 1 = Strongly disagree, 2 = Disagree, 3 = Neither agree nor disagree, 4 = Agree, 5 = Strongly agree.

### A.1 Training & Awareness (TA)

1. My firm provides specific training on cognitive biases (such as anchoring) that can affect audit judgments.
2. I have been taught concrete techniques for avoiding over-reliance on prior-year figures.
3. Training in my firm includes practice exercises with feedback, not only lectures.
4. (R) I have received little or no training on how initial reference points can distort judgment.
5. I feel well prepared to recognize when an anchor is influencing my conclusion.

### A.2 Rotation of Auditors (RA)

1. Personnel on my long-term engagements are rotated on a regular schedule.
2. Rotation in my firm is frequent enough to bring a "fresh look" to recurring clients.

3 When I join a continuing engagement, I form my own view before reading prior auditors' conclusions.

4 (R) The same individuals remain on my engagements for many years without change.

5 Rotation policies in my firm meaningfully reduce over-familiarity with clients.

**A.3 Use of Analytical Tools (AT)**

1 I routinely use data-analytics tools that flag anomalies in client data.

2 Visualization tools help me notice patterns that differ from prior-year expectations.

3 The analytical tools I use draw on data beyond the client's own ledgers.

4 (R) I rely mainly on prior-year workpapers rather than on current-period analytics.

5 The tools I use prompt me to investigate items that contradict my expectations.

**A.4 Structured Auditing Processes (SAP)**

1 My audit work follows standardized procedures and decision aids.

2 Procedures require me to evaluate current evidence before consulting the prior-year conclusion.

3 Documentation templates make my reasoning explicit and reviewable.

4 (R) My approach to recurring tasks is mostly informal and ad hoc.

5 The structure of my firm's methodology reduces reliance on mental shortcuts.

*Attention check 1:* To confirm you are reading carefully, please select "Disagree" for this item.

**A.5 Feedback & Reflection (FR)**

1 I receive feedback on the quality of my judgments, not only on meeting deadlines.

2 Post-engagement debriefs prompt me to reflect on judgment errors.

3 Feedback compares my conclusions against independent evidence.

4 (R) I rarely learn whether my professional judgments were accurate.

5 Structured reflection is a regular part of how I improve my work.

**A.6 Independent Reviews (IR)**

1 My work is reviewed by a qualified auditor who was not involved in forming the original judgment.

2 Reviewers in my firm independently re-examine key estimates.

3 The review process catches instances where I have under-adjusted from a prior-year figure.

4 (R) Reviews in my firm mostly confirm the preparer's conclusion without independent analysis.

5 Engagement-quality reviews are substantive rather than a formality.

**A.7 Regulatory & Professional Guidance (RPG)**

1 Regulatory standards (e.g., PCAOB) meaningfully shape how I exercise judgment.

2 Inspection risk motivates me to reconsider judgments more carefully.

3 Professional-skepticism requirements are actively reinforced in my firm.

4 (R) Regulatory guidance has little practical effect on my day-to-day judgments.

5 Standards require me to document independent support for estimates.

**A.8 Performance Metrics & Incentives (PMI)**

1 My performance evaluation rewards judgment quality, not only efficiency.

2 Incentives in my firm are aligned with audit quality.

3 I am recognized for exercising appropriate professional skepticism.

4 (R) Pressure to meet budgets discourages me from re-examining prior-year balances.

5 The way my work is evaluated encourages thorough, independent analysis.

*Attention check 2:* Please select "Agree" for this item to confirm your attention.

**A.9 Auditor Judgment Quality (AJQ) — mediator**

1 My audit judgments are based on careful evaluation of independent evidence.

2 I reach conclusions objectively, free of undue influence from management's explanations.

3 I deploy relevant knowledge and reasoning when forming difficult judgments.

4 (R) My conclusions often default to last year's position.

5 I am confident my judgments would withstand independent scrutiny.

**A.10 Audit Process Rigor (APR) — mediator**

1 The audit process I follow is thorough and consistently executed.

2 Key judgments are independently corroborated within the engagement.

3 The process enforces disciplined evaluation of evidence in a defined order.

4 (R) Execution on my engagements is often rushed or inconsistent.

5 The rigor of the process does not depend on any single individual.

**A.11 Reduction in Anchoring Bias (RAB) — dependent variable**

1 When auditing a recurring account, I adjust fully away from the prior-year balance when current evidence warrants.

2 I form an independent expectation before I look at management's estimate.

3 My final conclusions are driven by current evidence rather than by initial reference points.

4 (R) My estimates tend to stay close to the prior-year figure even when evidence suggests otherwise.

5 I actively seek evidence that contradicts my initial expectation.

**A.12 Demographic and control items**

Years of audit experience; primary role (internal/external); firm type (public accounting / private / regulatory / government); industry focus; exposure to long-term (multi-year) engagements (yes/no); highest credential (CPA/CIA/other); region.

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